

DASC7011 Statistical Inference for Data Science

Chapter 2 Method of Moments Estimation

Department of Statistics and Actuarial Science
Department of Computer Science
The University of Hong Kong

June 8, 2024

§2.1 Method-of-Moments Estimation

2.1.1 Moments

2.1.2 Sample moments

2.1.3 Method-of-moments estimation

§2.2 Statistical Inferences Based on MMEs

2.2.1 Statistical inferences on one mean

2.2.2 Statistical inferences on variances

2.2.3 Statistical inferences on two means

2.2.4 The Binomial test

2.2.5 The Jacque-Bera test

§2.1 Method-of-Moments Estimation

2.1.1. Moments and sample moments

- Let $X \sim F$ be a random variable with distribution F .
- The n -th **raw moment** (crude moment, moment about zero) of X (or F) is defined as (if exist):

$$\mu_n = \mathbb{E}(X^n), \quad n \geq 1.$$

- The first (order) raw moment $\mu_1 = \mathbb{E}(X)$ is usually referred to as the **mean** or **expectation** of X , and denoted as μ .
- **Hausdorff moment problem**: For a distribution of mass or probability on a bounded interval, the collection of all the moments uniquely determines the distribution.

Central moments

- Suppose $X \sim F$ has a finite mean $\mu < \infty$.
- The n -th **central moment** of X is defined as:

$$\sigma_n = \mathbb{E}(X - \mu)^n, \quad n \geq 2.$$

- The second (order) central moment σ_2 , if exists, is called the **variance** of X , and denoted as σ^2 .
- The non-negative square root of σ^2 , denoted as σ , is called the **standard deviation** of X .

Standardized moments: skewness

- Suppose $X \sim F$ has a finite variance $\sigma^2 < \infty$, and hence has a finite mean $\mu < \infty$.
- The n -th **standardized moment**:

$$\eta_n = \frac{\sigma_n}{\sigma^n}, \quad n \geq 3.$$

- The 3rd standardized moment η_3 is called the **skewness**.

$$\eta = \eta_3 = \frac{\mathbb{E}(X - \mu)^3}{\sigma^3} = \mathbb{E} \left(\frac{X - \mu}{\sigma} \right)^3.$$

- Skewness is a measure of the asymmetry of the distribution F . We say F is positively or negatively skewed if $\eta > 0$ or < 0 .
- If X is normal, then $\eta = 0$.

Standardized moments: kurtosis

- The 4th standardized moment η_4 is called the **kurtosis**.

$$\kappa = \eta_4 = \frac{\mathbb{E}(X - \mu)^4}{\sigma^4} = \mathbb{E} \left(\frac{X - \mu}{\sigma} \right)^4.$$

- Kurtosis is a measure of the “tailedness” of the distribution F .
- If X is normal, then $\kappa = 3$.
- $\kappa > 3$ indicates that the distribution F has *fatter* tail(s) than normal.
- We call $(\kappa - 3)$ the **excess kurtosis**.

Covariance and correlation

- When a random vector, say (X, Y) is considered, two more important moments are frequently studied.
- Suppose X has finite mean μ_x and variance σ_x^2 , and Y has finite mean μ_y and variance σ_y^2 .
- The *mixed 2nd central moment*

$$\sigma_{x,y} = \mathbb{E}[(X - \mu_x)(Y - \mu_y)]$$

is called the **covariance** between X and Y .

- The *mixed 2nd standardized moment*

$$\rho_{x,y} = \frac{\sigma_{x,y}}{\sigma_x \sigma_y}$$

is called the **correlation**, or **linear correlation coefficient**, between X and Y .

Sample moments

2.1.2. Sample moments

- Moments are expectations, or functions of expectations, of X , the **population**.
- Let $\mathbf{X} = \{X_1, \dots, X_n\}$ be a **sample** from a population $X \sim F$.
- **The principal statistical idea**: estimating the expectation $\mathbb{E}[f(X)]$ by the average

$$\overline{f(X)} = \frac{1}{n} \sum_{i=1}^n f(X_i), \quad (2.1)$$

where $f(\cdot)$ is a (general) function.

- Applying this *averaging algorithm* to moments, we obtain **sample moments**.

Sample moments

- **Sample raw moments:**

$$\hat{\mu}_k = \frac{1}{n} \sum_{i=1}^n X_i^k, \quad k = 1, 2, \dots$$

- **Sample central moments** when μ is known:

$$\hat{\delta}_k = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^k, \quad k = 2, 3, \dots$$

- **Sample central moments** when μ is unknown:

$$\hat{\sigma}_k = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu}_1)^k, \quad k = 2, 3, \dots$$

Sample mean and sample 2nd central moment

- We call the sample 1st raw moment

$$\bar{X}_n = \hat{\mu}_1 = \frac{1}{n} \sum_{i=1}^n X_i$$

the **sample mean** of X .

- The **sample 2nd central moment** of X is defined as

$$W_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2, \quad \mu \text{ is known,}$$
$$T_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2, \quad \mu \text{ is unknown,}$$

where the suffix n in $\bar{X}_n$ is dropped for simplicity.

Sample variance

- Generally, the population mean μ would be unknown.
- We call the *bias-corrected* sample 2nd central moment

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

the **sample variance** of X .

- The non-negative square root of S_n^2 ,

$$S_n = \left(\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \right)^{1/2},$$

is called the **sample standard deviation**.

Sample skewness and sample kurtosis

- **Sample skewness:**

$$\hat{\eta} = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^3}{\hat{\sigma}_2^{3/2}} \quad \text{or} \quad \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^3}{S^3}.$$

- **Sample kurtosis:**

$$\hat{\kappa} = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^4}{\hat{\sigma}_2^2} \quad \text{or} \quad \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^4}{S^4}.$$

- **Remark:** There are several slightly different definitions of sample skewness and sample kurtosis in literature, regarding the unbiasedness and/or efficiency.

Sample covariance and sample correlation

- Let $\{(X_i, Y_i) : 1 \leq i \leq n\}$ be a random sample from a bivariate population (X, Y) , $\bar{X}$ and $\bar{Y}$ be the sample means, S_x^2 and S_y^2 be the sample variances, respectively.
- The **sample covariance** is defined as

$$S_{x,y} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}).$$

- The **sample correlation** is defined as

$$\hat{\rho}_{x,y} = \frac{S_{x,y}}{S_x S_y}.$$

It is also known as the **Pearson correlation coefficient** in literature, and denoted as $r_{x,y}$.

Sample moments in R

- **Example 2.1:** We use two generated samples with size $n = 100$, $\{X_i\} \stackrel{i.i.d.}{\sim} N(5, 4)$ and $\{Y_i\} \stackrel{i.i.d.}{\sim} t(4)$, for illustration.
- Built-in functions of sample moments in R.

Moment	$\bar{X}$	S_x^2	S_x	$S_{x,y}$	$r_{x,y}$	$\hat{\eta}_x$	$\hat{\kappa}_x$ -3
R Function	mean	var	sd	cov	cor	skewness*	kurtosis*
Estimate	4.596	4.159	2.039	0.052	0.020	-0.039	-0.459

* In R package **e1071**.

- You can calculate them using your own functions/R expressions, to check the definitions of built-in formulas.
- In my attached program, both build-in and self-defined functions give the same estimates (estimated results). □

Method-of-Moments Estimation (MME)

2.1.3. Method-of-Moments Estimation (MME)

- It is noticeable that the skewness is NOT an *original moment* (an expectation) by definition, but a (rational) function of two moments σ_3 and σ_2 . In the meantime, we defined the sample skewness as (the same) function of corresponding sample moments $\hat{\sigma}_3$ and S^2 (or $\hat{\sigma}_2$).
- This is an immediate generalization of the principal statical idea, or the averaging algorithm.
- The idea behind: there exist certain mathematic relationship(s) between the target character (skewness, to be estimated) and moments.
- This leads to the **Method-of-Moments Estimation (MME)**, or the MME algorithm.

Method-of-Moments Estimation (MME)

- Let $\mathbf{X} = \{X_1, \dots, X_n\}$ be a sample from a parametric (not a must) model $X \sim \mathcal{M}(\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ are unknown parameter(s).
- The moments m_k 's, if exist, will generally depend on the parameter(s) $\boldsymbol{\theta}$. If possible, they might be expressed as functions of $\boldsymbol{\theta}$, either explicit or implicit,

$$m_k = f_k(\boldsymbol{\theta}), \quad 1 \leq k \leq p,$$

for some $p \geq 1$. These functions are also known as *moment conditions* or *moment equations*.

- Suppose we can solve these moment equations for parameter(s) to obtain

$$\boldsymbol{\theta} = g(m_1, m_2, \dots, m_p).$$

Method-of-Moments Estimation (MME)

- Replacing m_k 's in the solution with corresponding sample moments $\hat{m}_k$'s, we obtain the **method-of-moments estimate(s)** (**MME**) of parameter(s):

$$\hat{\theta} = g(\hat{m}_1, \hat{m}_2, \dots, \hat{m}_p). \quad (2.2)$$

- By definition, all sample moments in subsection 2.1.2 are MMEs of corresponding population moments.
- Remark:** moment equations are not always solvable for parameters θ (either existence or uniqueness). The **generalized method-of-moments estimation (GMME)** method provides a generalization of MME for this purpose.
 - GMME applies a statistical algorithm called *least-squares* to the estimation, and will be introduced in the next chapter.

Example 2.2: MMEs are not unique

- **Example 2.2:** Let $\mathbf{X} = \{X_1, \dots, X_n\}$ be a random sample from an exponential distribution (population) with unknown rate λ . The pdf is $f(x) = \lambda e^{-\lambda x}$ for all $x > 0$.
- $\hat{\lambda}_1 = \bar{X}^{-1}$ is one possible MME of λ since $\mathbb{E}(X) = 1/\lambda$.
- $\hat{\lambda}_2 = S^{-1}$ is another valid MME of λ since $\text{Var}(X) = 1/\lambda^2$, where S is the sample standard deviation of $\mathbf{X}$.
- **Remarks:** MMEs of parameters are not unique. Generally, we prefer using lower order moment(s) in finding MME(s) due to its/their better properties (convergence under weaker conditions).
 - We prefer $\hat{\lambda}_1 = \bar{X}^{-1}$ in this example. □

Example 2.3: MME for AR models

- **Example 2.3:** Consider a sample $\{X_t : 1 \leq t \leq T\}$ from an AR(1) model

$$X_t = \phi_0 + \phi_1 X_{t-1} + a_t, \quad \{a_t\} \stackrel{i.i.d.}{\sim} (0, \sigma_a^2),$$

where ϕ_0 , $-1 < \phi_1 < 1$ and $\sigma_a^2 > 0$ are three parameters to be estimated.

- Under condition $-1 < \phi_1 < 1$, the model is *second order stationary*: the 1st and 2nd order moments are stationary (invariant) against time drift/transformation.

Example 2.3: MME for AR models

- Three moment conditions are:

$$\mu_x = \mathbb{E}(X_t) = \frac{\phi_0}{1 - \phi_1},$$

$$\rho_1 = \text{Corr}(X_t, X_{t-1}) = \phi_1,$$

$$\gamma_0 = \text{Var}(X_t) = \frac{\sigma_a^2}{1 - \phi_1^2}.$$

- Solving these, we have

$$\phi_1 = \rho_1, \quad \phi_0 = \mu_x(1 - \rho_1), \quad \sigma_a^2 = \gamma_0(1 - \rho_1^2).$$

Example 2.3: MME for AR models

- Let $\bar{X}$ and S^2 be the sample mean and sample variance of $\{X_t\}$, respectively. Moreover, defined the *sample autocorrelation* at lag 1 as

$$\begin{aligned} r_1 &= \widehat{\text{Corr}}(X_t, X_{t-1}) \\ &= \frac{\sum_{t=2}^T (X_t - \bar{X})(X_{t-1} - \bar{X})}{\sum_{t=1}^T (X_t - \bar{X})^2}. \end{aligned}$$

- Then, the MMEs are

$$\hat{\phi}_1 = r_1, \quad \hat{\phi}_0 = \bar{X}(1 - r_1), \quad \hat{\sigma}_a^2 = S^2(1 - r^2).$$

□

Example 2.4: MME for Gamma distribution

- **Example 2.4:** Let $\{X_i : 1 \leq i \leq n\}$ be a random sample from a Gamma population $X \sim \Gamma(\alpha, \beta)$ with pdf

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0,$$

where $\alpha > 0$ and $\beta > 0$ are the *shape* and *rate* parameters.

- There is no explicit formula for the maximum likelihood estimator (MLE, will be introduced in Chapter 4) due to the complexity of Gamma function $\Gamma(\alpha)$.
- On the other hand, the MMEs can be easily obtained.

Example 2.4: MME for Gamma distribution

- The Gamma function is defined as

$$\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} e^{-t} dt, \quad \alpha > 0,$$

and has property $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$.

- Then,

$$\begin{aligned}\mathbb{E}(X) &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^{\infty} x^{\alpha-1} e^{-\beta x} x dx \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^{\infty} (\beta x)^\alpha \beta^{-\alpha-1} e^{-\beta x} d(\beta x) \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)} \beta^{-\alpha-1} \Gamma(\alpha + 1) = \frac{\alpha}{\beta}.\end{aligned}$$

Example 2.4: MME for Gamma distribution

- Similarly, it can be found that

$$\text{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = \frac{\alpha}{\beta^2}.$$

- These solve for

$$\alpha = \frac{[\mathbb{E}(X)]^2}{\text{Var}(X)}, \quad \beta = \frac{\mathbb{E}(X)}{\text{Var}(X)}.$$

- Therefore, the MMES are

$$\hat{\alpha} = \frac{(\bar{X})^2}{S^2}, \quad \hat{\beta} = \frac{\bar{X}}{S^2},$$

where $\bar{X}$ and S^2 are the sample mean and sample variance of $\{X_i\}$, respectively. $\square$

Example 2.5: MME for rate/probability

- **Example 2.5:** Let X denote the age of a general people in Hong Kong. We are interested in the proportion (rate/probability) of elder people, say, $p = \mathbb{P}(X \geq 65)$.
- Notice that $p = \mathbb{E}[1(X \geq 65)]$, where $1(A)$ stands for the indicator function of event A . The MME can be applied to estimate this rate.
- Suppose a random sample $\{X_i : 1 \leq i \leq n\}$ is drawn from the population X . Then

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n 1(X_i \geq 65).$$



§2.2 Statistical Inferences Based on MMEs

- Properties (mean, variance, bias and distribution) of sample moments, as estimates of population moments, are studied.
- Let $0 < \alpha < 1$ be a constant. Through out this section, all confidence intervals are constructed at the $(1 - \alpha)$ -level of confidence, and all tests are conducted at the α -level of significance.

Properties of sample mean

2.2.1. Statistical inference on one mean.

- Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean of a *random sample* $\{X_i : 1 \leq i \leq n\}$ from a population $X \sim F$ with finite mean μ and variance σ^2 .
- $\mathbb{E}(\bar{X}_n) = \mu$ and $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$. Hence, $\bar{X}_n$ is an **unbiased** and **consistent** estimator of μ .
- If $X \sim N(\mu, \sigma^2)$ is normal, then

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right). \quad (2.3)$$

- For a general population F , by CLT, $\bar{X}$ is **asymptotically normal**,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \rightsquigarrow N(0, 1) \quad \text{for large } n. \quad (2.4)$$

CIs of one population mean

(a) σ^2 is known.

(i) If $X \sim N(\mu, \sigma^2)$ is normal, by (2.3),

$$\begin{aligned} \mathbb{P} \left(-Z_{\alpha/2} < \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} < Z_{\alpha/2} \right) &= 1 - \alpha, \\ \Leftrightarrow \mathbb{P} \left(\bar{X}_n - Z_{\alpha/2}\sigma/\sqrt{n} < \mu < \bar{X}_n + Z_{\alpha/2}\sigma/\sqrt{n} \right) &= 1 - \alpha. \end{aligned}$$

Therefore, the confidence of μ is

$$(\bar{X}_n - Z_{\alpha/2}\sigma/\sqrt{n}, \bar{X}_n + Z_{\alpha/2}\sigma/\sqrt{n}). \quad (2.5)$$

Cf. Example 1.2.

(ii) For general F , by the asymptotical distribution in Eq. (2.4), the CI in (2.5) is approximately valid for large n .

CIs of one population mean

(b) σ^2 is unknown.

- (i) If $X \sim N(\mu, \sigma^2)$ is normal, then $\bar{X}_n$ and S_n^2 are independent. Moreover,

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1), \quad \text{and} \quad \frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \sim t(n-1). \quad (2.6)$$

Therefore, the exact confidence interval of μ becomes

$$(\bar{X}_n - t_{\alpha/2}(n-1)S_n/\sqrt{n}, \bar{X}_n + t_{\alpha/2}(n-1)S_n/\sqrt{n}). \quad (2.7)$$

- (ii) For general F , by the asymptotical distribution in (2.4), and the fact that $t(n-1)$ is very closed to $N(0,1)$ when n is large, the approximate CI is

$$(\bar{X}_n - Z_{\alpha/2}S_n/\sqrt{n}, \bar{X}_n + Z_{\alpha/2}S_n/\sqrt{n}). \quad (2.8)$$

Testing one population mean

- For illustration, consider testing $H_0 : \mu = \mu_0$ vs $H_a : \mu \neq \mu_0$ for some constant μ_0 .

(a) σ^2 is known.

- (i) When $X \sim N(\mu, \sigma^2)$ is normal, the test statistic is

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1) \quad \text{under } H_0.$$

Therefore, the rejection region is

$$R_\alpha = \{|Z| > Z_{\alpha/2}\},$$

where $Z_{\alpha/2}$ is the upper quantile of standard normal at the $\frac{\alpha}{2}$ level.

- (ii) For general F , the above results are approximately true for large n .

Testing one population mean

(b) σ^2 is unknown.

(i) When $X \sim N(\mu, \sigma^2)$ is normal, the test statistic is

$$T = \frac{\bar{X}_n - \mu_0}{S_n / \sqrt{n}} \sim t(n-1) \quad \text{under } H_0,$$

where $t(n-1)$ is the student- t distribution with $(n-1)$ degrees of freedom. Therefore, the rejection region is

$$R_\alpha = \{|T| > t_{\alpha/2}(n-1)\},$$

where $t_{\alpha/2}(n-1)$ is the upper quantile of $t(n-1)$ distribution at the $\frac{\alpha}{2}$ level.

(ii) For general F , the approximate (for large n) rejection region is

$$R_\alpha = \{|T| > Z_{\alpha/2}\}.$$

Example 2.6: Inferences on one mean

- **Example 2.6.** Consider the sample $\{x_i : 1 \leq i \leq 100\}$ generated from a normal $N(5, 2^2)$ population in Example 2.1, for which we have $\bar{X} = 4.596$ and $S = 2.039$. We are constructing the 95% confidence intervals, and testing $H_0 : \mu = 4$ at the 5% significance level.
- (i) Assuming $\sigma = 2$ is known. There is no build-in R function for this case.
 - Using Formula (2.5), the 95% CI is calculated to be (4.204, 4.988).
 - The test statistic is $z = 2.982$. H_0 is rejected since $|z| > Z_{0.025} = 1.96$. Moreover, the p -value is (where $Z \sim N(0, 1)$)

$$p = 2 \times \mathbb{P}(|Z| > |z|) = 0.00286.$$

Example 2.6: Inferences on one mean

(ii) Manual calculation assuming σ is unknown.

- Using Formula (2.7), the 95% CI is calculated to be (4.192, 5.001).
- The test statistic is $t = 2.924$. H_0 is rejected since $|z| > t_{0.025}(99) = 1.984$. The p -value is (where $T \sim t(99)$)

$$p = 2 \times \mathbb{P}(|T| > |t|) = 0.00428.$$

(iii) The build-in R function `t.test()` can be used to do the inferences in case (ii) including both the test and CI.

```
# (iii) Using t.test()
t.test(x, alternative = "two.sided", mu=mu0)
t.test(x, alternative = "less", mu=mu0)
t.test(x, alternative = "greater", mu=mu0)
```

Example 2.6: Inferences on one mean

- The first line tests H_0 against $H_1 : \mu \neq 4$. It gives the same results as in (ii).
- The last two lines do the one-sided test. Refer to the following output.

```
> t.test(x, alternative = "greater", mu=mu0)
      One Sample t-test
data:  x
t = 2.9245, df = 99, p-value = 0.002139
alternative hypothesis: true mean is greater than 4
95 percent confidence interval:
 4.257796      Inf
sample estimates:
mean of x
 4.596415
```



Properties of MMEs of variance

2.2.2. Statistical inference on variances.

- The sample variance S_n^2 is an unbiased estimator of the population variance σ^2 , and

$$\text{Var}(S_n^2) = \frac{1}{n} \left(\sigma_4 - \frac{n-3}{n-1} \sigma^4 \right).$$

Notice that $\lim_{n \rightarrow \infty} \text{Var}(S_n^2) = 0$. Hence, $\{S_n^2\}$ is consistent.

- If the population $X \sim N(\mu, \sigma^2)$ is normal, then $\bar{X}_n$ and S_n^2 are independent, and

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1). \quad (2.9)$$

(We repeat this once again.)

Properties of MMEs of variance

- T_n^2 is biased with

$$\mathbb{E}(T_n^2) = \frac{(n-1)\sigma^2}{n}, \quad \text{bias}(T_n^2) = -\frac{\sigma^2}{n}, \quad n \in \mathbb{N}.$$

- The mean squared error (MSE) of T_n^2 is

$$\text{mse}(T_n^2) = \frac{1}{n^3} \left[(n-1)^2 \sigma_4 - (n^2 - 5n + 3) \sigma^4 \right], \quad n \in \mathbb{N}.$$

So, $\{T_n^2\}$ is consistent.

- When μ is known, $\mathbb{E}(W_n^2) = \sigma^2$, so W_n^2 is unbiased. Moreover,

$$\text{Var}(W_n^2) = \frac{1}{n}(\sigma_4 - \sigma^4), \quad n \in \mathbb{N}.$$

So $\{W_n^2\}$ is consistent.

Properties of MMEs of variance

- It is noticeable that $\text{Var}(W_n^2) < \text{Var}(S_n^2)$ for all $n \geq 3$, and hence W_n^2 is preferred when μ is known.
- W_n^2 and S_n^2 are asymptotically equivalent since

$$\frac{\text{Var}(W_n^2)}{\text{Var}(S_n^2)} \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

- There is no simple, general relationship between $\text{mse}(T_n^2)$ and $\text{mse}(S_n^2)$ or between $\text{mse}(T_n^2)$ and $\text{mse}(W_n^2)$, but the asymptotic relationship is simple. As $n \rightarrow \infty$,

$$\frac{\text{mse}(T_n^2)}{\text{mse}(W_n^2)} \rightarrow 1, \quad \text{and} \quad \frac{\text{mse}(T_n^2)}{\text{mse}(S_n^2)} \rightarrow 1.$$

Properties of MMEs of variance

- Comparisons become interesting if the population is normal, $X \sim N(\mu, \sigma^2)$.
- Mean squared errors of S_n^2 and T_n^2 .
 - $\text{mse}(T_n^2) = \frac{2n-1}{n^2} \sigma^4$.
 - $\text{mse}(S_n^2) = \frac{2}{n-1} \sigma^4$.
 - $\text{mse}(T_n^2) < \text{mse}(S_n^2)$ for $n \geq 2$.
- Mean squared errors of W_n^2 and T_n^2 .
 - $\text{mse}(W_n^2) = \frac{2}{n} \sigma^4$.
 - $\text{mse}(T_n^2) < \text{mse}(W_n^2)$ for $n \geq 2$.

Properties of MMEs of variance

- When $X \sim N(\mu, \sigma^2)$ is normal with known μ ,

$$\frac{nW_n^2}{\sigma^2} \sim \chi^2(n), \quad n \geq 1. \quad (2.10)$$

- Asymptotically (for large n),

$$\frac{S_n^2 - \sigma^2}{\sqrt{2\sigma^2/\sqrt{n-1}}} \approx \frac{T_n^2 - \sigma^2}{\sqrt{2\sigma^2/\sqrt{n}}} \approx \frac{W_n^2 - \sigma^2}{\sqrt{2\sigma^2/\sqrt{n}}} \dot{\sim} N(0, 1). \quad (2.11)$$

- Confidence intervals are not difficult to be obtained based on distributional properties in Eq.s (2.9) to (2.11).

Properties of sample standard deviation

- Assume $X \sim N(\mu, \sigma^2)$ is normal.
- Let

$$a_n = \frac{\sqrt{2}\Gamma(\frac{n+1}{2})}{\sqrt{n}\Gamma(\frac{n}{2})}, \quad n \geq 1.$$

Then, $0 < a_n < 1$ and $a_n \uparrow 1$ as $n \rightarrow \infty$.

- Means, biases, variances and mean squared errors of S and W are summarized in the following table.

	Mean	Bias	Variance	MSE
S	$a_{n-1}\sigma$	$(a_{n-1} - 1)\sigma$	$(1 - a_{n-1}^2)\sigma^2$	$2(1 - a_{n-1})\sigma^2$
W	$a_n\sigma$	$(a_n - 1)\sigma$	$(1 - a_n^2)\sigma^2$	$2(1 - a_n)\sigma^2$

Testing one variance

- Assume μ is unknown. Consider testing $H_0 : \sigma^2 \leq \sigma_0^2$ against $H_a : \sigma^2 > \sigma_0^2$ for some constant σ_0^2 .

- (a) If $X \sim N(\mu, \sigma^2)$ is normal, by (2.9), the test statistic is

$$\chi^2 = \frac{(n-1)S_n^2}{\sigma_0^2} \sim \chi^2(n-1) \quad \text{if } \sigma^2 = \sigma_0^2.$$

H_0 is rejected at the α -level of significance if $\chi^2 > \chi_\alpha^2(n-1)$, the α -th upper quantile of the $\chi^2(n-1)$ distribution.

- (b) For general F and large n , define

$$Z = \frac{S_n^2 - \sigma_0^2}{\sqrt{2\sigma_0^2/\sqrt{n-1}}} \dot{\sim} N(0, 1) \quad \text{if } \sigma^2 = \sigma_0^2.$$

Reject H_0 if $Z > Z_\alpha$. (Seldom used.)

Comparing two population variances

- Let $\{X_i : 1 \leq i \leq n_1\}$ and $\{Y_j : 1 \leq j \leq n_2\}$ be random samples from populations $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$, respectively. Assume both means are unknown.
- Testing $H_0 : \sigma_x \leq \sigma_y$ against $H_a : \sigma_x > \sigma_y$.
- By (2.9), the test statistic:

$$F = \frac{S_x^2}{S_y^2} = \frac{\frac{(n_1-1)S_x^2}{\sigma_x^2} / (n_1 - 1)}{\frac{(n_2-1)S_y^2}{\sigma_y^2} / (n_2 - 1)} \sim F(n_1 - 1, n_2 - 1) \quad \text{if } \sigma_x^2 = \sigma_y^2.$$

- Reject H_0 in favor of H_a if $F > F_\alpha(n_1 - 1, n_2 - 1)$, the α -th upper quantile of the F -distributions with degrees of freedom $(n_1 - 1)$ and $(n_2 - 1)$.

Example 2.7. Inferences on variances

- **Example 2.7.** Consider two samples: $\{X_i : 1 \leq i \leq 100\}$ from $X \sim N(5, 4)$ as in Example 2.1, and $\{Y_j : 1 \leq j \leq 200\}$ from $Y \sim N(0, 6)$.
- A different seed is used in the *random number generator* for Y to ensure the independence between two samples.

```
# Generate samples from  $Y \sim N(0, 6)$   
n2 <- 200  
set.seed(2024)  
y <- rnorm(n2, 0, sqrt(6))
```

```
> cor(x, y[1:100])  
[1] -0.07510114  
> cor(x, y[101:200])  
[1] -0.05394609
```

Example 2.7. Inferences on variances

(i) Test $H_0 : \sigma_x^2 = 4$ vs $H_a : \sigma_x^2 \neq 4$.

- The R function `EnvStats::varTest`: *One-Sample Chi-Squared Test on Variance*, is used.

```
> varTest(x, alternative = "two.sided", sigma.squared = 4)
Results of Hypothesis Test
-----
Null Hypothesis:                variance = 4
Alternative Hypothesis:         True variance is not equal to 4
Test Name:                      Chi-Squared Test on Variance
Estimated Parameter(s):         variance = 4.159123
Data:                           x
Test Statistic:                 Chi-Squared = 102.9383
Test Statistic Parameter:      df = 99
P-value:                        0.7463159
95% Confidence Interval:        LCL = 3.206251
                                UCL = 5.612692
```

Example 2.7. Inferences on variances

(ii) Test $H_0 : \sigma_x^2 \geq \sigma_y^2$ vs $H_a : \sigma_x^2 < \sigma_y^2$.

- The R function `var.test`: *F Test to Compare Two Variances*, is used.

```
> var.test(x, y, ratio = 1, alternative = "less")
```

F test to compare two variances

data: x and y

F = 0.65959, num df = 99, denom df = 199, p-value =
0.01051

alternative hypothesis: true ratio of variances is less than 1
95 percent confidence interval:

0.0000000 0.8861131

sample estimates:

ratio of variances
0.6595878



Statistical inference on two means.

2.2.3. Statistical inference on two means.

- Let $\{X_i : 1 \leq i \leq n_1\}$ and $\{Y_j : 1 \leq j \leq n_2\}$ be random samples from populations $X \sim F_x(\mu_x, \sigma_x^2)$ and $Y \sim F_y(\mu_y, \sigma_y^2)$, respectively. Assume both variances are unknown.
- Denote the sample means, sample variances, and sample covariance as $\bar{X}$, $\bar{Y}$, S_x^2 , S_y^2 , and $S_{x,y}$, respectively.
- Practically it is very common to do statistical inference on both population means μ_x and μ_y , e.g., in clinical trials.
- We consider testing $H_0 : \mu_x \geq \mu_y$ against $H_a : \mu_x < \mu_y$ for illustration.
- The test depends on whether $\sigma_x^2 = \sigma_y^2$ or not, and on whether X and Y are independent.

The two-samples t -test

- (a) X and Y are independently normal with equal variance $\sigma_x^2 = \sigma_y^2 = \sigma^2$.

- The difference in sample means $\bar{X} - \bar{Y}$ is normal with variance $\frac{\sigma_x^2}{n_1} + \frac{\sigma_y^2}{n_2} = \sigma^2(\frac{1}{n_1} + \frac{1}{n_2})$, in which σ^2 can be estimated by the **pooled estimator**

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} \left(\sum_{i=1}^{n_1} (X_i - \bar{X})^2 + \sum_{j=1}^{n_2} (Y_j - \bar{Y})^2 \right).$$

- Define the test statistic as

$$t = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2) \quad \text{under } H_0. \quad (2.12)$$

- Reject H_0 if $t < -t_\alpha(n_1 + n_2 - 2)$.

The Welch's t -test

(b) X and Y are independently normal with unequal $\sigma_x^2 \neq \sigma_y^2$.

- $\bar{X} - \bar{Y}$ is normal, and its variance is estimated by $\frac{S_x^2}{n_1} + \frac{S_y^2}{n_2}$.
- The t -test statistic is

$$t = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_x^2}{n_1} + \frac{S_y^2}{n_2}}} \sim t(v) \quad \text{under } H_0, \quad (2.13)$$

where the degrees of freedom is a real number instead of an integer,

$$v \approx \frac{\left(\frac{S_x^2}{n_1} + \frac{S_y^2}{n_2}\right)^2}{\frac{S_x^4}{n_1^2(n_1-1)} + \frac{S_y^4}{n_2^2(n_2-1)}}.$$

- Reject H_0 if $t < -t_\alpha(v)$.

Two-samples t -test or Welch's t -test?

- Welch's t -test and Student's t -test give identical results when the two samples have equal variances and sample sizes. The power of Welch's t -test comes close to that of Student's t -test, even when the population variances are equal and sample sizes are *balanced*.¹
- Welch's t -test is more robust than two-samples Student's t -test and maintains type I error rates close to nominal for unequal variances and for unequal sample sizes under normality.
- It is *not recommended* to pre-test for equal variances and then choose between two-samples t -test or Welch's t -test.²
- When sample sizes are large, both tests are approximately valid for non-normal populations.

¹Wikipedia

²Zimmerman, D.W. (2004). A note on preliminary tests of equality of variances. *British Journal of Mathematical and Statistical Psychology*, **57** (Pt 1): 173-181.

The paired-samples t -test

(c) If $\{(X_i, Y_i) : 1 \leq i \leq n\}$'s are paired observations.

- This arises when we are investigating the impacts of different treatments (conditions) on the same (batch of) individuals.
- Assumption: $\{D_i \triangleq X_i - Y_i : 1 \leq i \leq n\}$ are i.i.d. normal $N(\mu_d, \sigma_d^2)$.
- Testing $H_0 : \mu_d = 0$ vs $H_a : \mu_d \neq 0$ assuming σ_d^2 is unknown.
- The problem reduces to testing hypothesis on one mean using a single sample $\{D_i\}$. The one-sample t -test is applied. Refer to subsection 2.2.1.
- We also call this the **paired-samples t -test**.

Example 2.8. Inferences on two means

- **Example 2.8.** Samples $\{X_i\}$ and $\{Y_j\}$ in Example 2.7 will be used for cases (a) and (b), the independent samples t -tests assuming equal variance or not.
- A pair of samples $\{X_i, Y_i) : 1 \leq i \leq 100\}$ are generated for case (c), the paired t -test. X and Y are not independent.

```
> set.seed(7011)
> X <- rt(n,5)
> Y <- runif(n,-0.8,1)
> cor(X,Y)
[1] -0.09167719
```

- The R function `t.test` can be applied for all 3 cases.

Example 2.8. Inferences on two means

(a) Two-samples t -test assuming equal variance.

```
> t.test(x, y, alternative = "two.sided", var.equal = T)
```

Two Sample t-test

data: x and y

$t = 15.67$, $df = 298$, $p\text{-value} < 2.2e-16$

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

3.968702 5.108677

sample estimates:

mean of x mean of y

4.59641471 0.05772509

Example 2.8. Inferences on two means

(b) Welch's t -test assuming unequal variance.

```
> t.test(x, y, alternative = "two.sided", var.equal = F)
```

```
      welch Two Sample t-test
```

```
data:  x and y
```

```
t = 16.785, df = 237.96, p-value < 2.2e-16
```

```
alternative hypothesis: true difference in means is not  
equal to 0
```

```
95 percent confidence interval:
```

```
 4.005994 5.071385
```

```
sample estimates:
```

```
mean of x  mean of y
```

```
4.59641471 0.05772509
```

Example 2.8. Inferences on two means

(c) Paired samples t -test.

```
> t.test(X, Y, alternative = "two.sided", paired = T)
```

Paired t-test

data: X and Y

$t = -2.5966$, $df = 99$, $p\text{-value} = 0.01085$

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-0.61160806 -0.08176225

sample estimates:

mean of the differences

-0.3466852



2.2.4. The Binomial test.

- If the population distribution $F = \text{Bin}(1, p)$, the Bernoulli (bi-point) distribution, the MME of the success or failure rate p is the sample proportion (also the sample mean)

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i.$$

- Define $Y_n = \sum_{i=1}^n X_i$, then Y_n is Binomial $\text{Bin}(n, p)$, and

$$\mathbb{E}(Y_n) = np, \quad \text{Var}(Y_n) = np(1 - p).$$

- Since $\hat{p} = \frac{Y_n}{n}$,

$$\mathbb{E}(\hat{p}) = p, \quad \text{Var}(\hat{p}) = p(1 - p)/n.$$

Hence, $\hat{p}$ is unbiased and consistent.

The exact test and CI

- The exact distribution of Y_n is usually used to do the *exact test* on p , especially when n is not too large. Such a test is referred to as the **Binomial test**.
- Suppose we want to test $H_0 : p = p_0$ against $H_a : p \neq p_0$ for some $0 < p_0 < 1$.
- Let

$$y_1 = \max\{k : \mathbb{P}(Y_n \leq k | H_0) \leq \alpha/2\},$$
$$y_2 = \min\{k : \mathbb{P}(Y_n > k | H_0) \leq \alpha/2\}.$$

Then, H_0 is rejected if the observed $y_n \leq y_1$ or $\geq y_2$.

- By the duality between hypothesis test and confidence interval, the $(1 - \alpha)$ -th exact CI of p is the collection of all p_0 's such that we fail to reject the null hypothesis $H_0 : p = p_0$.

The approximate test

- When the sample size n is large ($n \geq 40$ is suggested), by CLT,

$$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \sim N(0, 1).$$

This asymptotical distribution is usually used to do approximate inferences on rate p , especially hypothesis tests.

- To test $H_0 : p = p_0$ against $H_a : p \neq p_0$, define

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0, 1) \quad \text{under } H_0.$$

H_0 is rejected at the α -level of significance if $|Z_0| > Z_{\alpha/2}$.

The approximate CI

- By the approximate distribution of Z , for any $0 < \alpha < 1$,

$$|Z| < Z_{\alpha/2} \iff (\hat{p} - p)^2 < c_{\alpha}p(1 - p),$$

where $c_{\alpha} = \frac{Z_{\alpha/2}^2}{n}$.

- Solving this inequality for p gives the analytical formula of the approximate CI $(\hat{p}_1, \hat{p}_2)$, where $\hat{p}_1 < \hat{p}_2$ are two roots of the quadratic equation $(\hat{p} - p)^2 = c_{\alpha}p(1 - p)$, i.e.,

$$\hat{p}_{1,2} = \frac{2\hat{p} + c_{\alpha} \mp \sqrt{c_{\alpha}^2 + 4c_{\alpha}\hat{p}(1 - \hat{p})}}{2(1 + c_{\alpha})}.$$

- Numerically, the approximate CI of p is usually constructed by the duality between hypothesis test and confidence interval.

Example 2.9. Binomial test

- **Example 2.9.** Consider a random sample of size $n = 50$ from a $\text{Bin}(1, 0.4)$ population, generated by the following DGP.

```
n <- 50
set.seed(7011)
X <- rbinom(n, 1, 0.4)
Y <- sum(X)           # Number of successes
X1 <- c(Y, n-Y)       # Numbers of successes/failures
p0 <- 0.5
```

- The number of successes is defined as Y , and $X1$ is a vector (of length 2) of both numbers of successes and failures.
- R function `binom.test()` provides two equivalent ways to do the test using Y and $X1$, respectively.
- We are testing $H_0 : p = p_0 = 0.5$ against $H_0 : p \neq p_0$.

Example 2.9. Binomial test

- R commands (two ways) are

```
binom.test(x = Y, n = n, p = p0)  
binom.test(x = x1, p = p0)
```

- Both ways give exactly the same numerical results.

```
data: Y and n          or          x1  
number of successes = 16, number of trials = 50,  
p-value = 0.01535  
alternative hypothesis: true probability of success is not  
equal to 0.5  
95 percent confidence interval:  
 0.1952042 0.4669938  
sample estimates:  
probability of success 0.32
```

Example 2.9. Binomial test

- The approximate test and corresponding analytical CI are calculated as follows.

```
> # Approximate Z-test
> Z <- (mean(X)-p0)/sqrt(p0*(1-p0)/n); Z
[1] -2.545584
> p.val <- pnorm(Z)*2; p.val
[1] 0.0109095
> # Approximate and analytic CI
> c <- qnorm(0.025)^2/n
> p.hat <- mean(X)
> p1 <- (2*p.hat+c-sqrt(c^2+4*c*p.hat*(1-p.hat)))/(2*(1+c))
> p2 <- (2*p.hat+c+sqrt(c^2+4*c*p.hat*(1-p.hat)))/(2*(1+c))
> c(p1, p2)
[1] 0.2075822 0.4581030
```

- The approximations are not too bad since $n = 50 > 40$. □

Properties of sample skewness and kurtosis

2.2.5. The Jarque-Bera test.

- There is a very popular *normality test* in econometrics called the **Jarque-Bera test**. It mainly uses the following properties of sample skewness and sample kurtosis from normal populations.
- Assume $X \sim N(\mu, \sigma^2)$ is normal. For large n ,

$$\hat{\eta} \dot{\sim} N\left(0, \frac{6}{n}\right). \quad (2.14)$$

- Assume $X \sim N(\mu, \sigma^2)$ is normal. For large n ,

$$\hat{\kappa} \dot{\sim} N\left(3, \frac{24}{n}\right). \quad (2.15)$$

The Jarque-Bera test

- Normality test: “ $H_0 : X$ is normal” vs “ $H_a : \text{not } H_0$ ”.
- The Jarque-Bera test statistics is defined as

$$JB = n \left(\frac{\hat{\eta}^2}{6} + \frac{(\hat{\kappa} - 3)^2}{24} \right). \quad (2.16)$$

- Under H_0 , $JB \sim \chi^2(2)$.
- Reject H_0 if $JB > \chi^2_{\alpha}(2)$.
- Inferences *purely* on higher order moments such as skewness and kurtosis are very seldom seen.

Example 2.10. Jarque-Bera test

- **Example 2.10.** Jarque-Bera test.

```
> require(tseries)
> x <- rnorm(100, 0, 1)
> jarque.bera.test(x)
```

Jarque Bera Test

```
data: x
X-squared = 0.32747, df = 2, p-value = 0.849
```

```
> y <- rcauchy(100, 0, 1)
> jarque.bera.test(y)
```

Jarque Bera Test

```
data: y
X-squared = 28840, df = 2, p-value < 2.2e-16
```

